# SUNGJAE AHN

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## RESEARCH INTERESTS

My conception of a fully realized AI is an agent that, as it accumulates experience, becomes better at predicting the consequences of its actions, learns more quickly which actions are effective in achieving its goals, and becomes better at determining how to modify its own behavior or structure—all to adapt to a world bigger than itself. To this end, I am interested in **(Continual) Reinforcement Learning**, **Exploration in RL**, and **Meta-learning for RL**.

## EDUCATION

**Kyung Hee University (KHU)** Expected Graduation: Aug. 2027
B.S. in Computer Science and Engineering
GPA: 4.31/4.5 (3.88/4.0); Upper-division: 4.48/4.5 (4.0/4.0)
Leave of absence for military service: Feb. 2022 – Nov. 2023

## RESEARCH EXPERIENCE

**AI & Robotics Lab, KHU** Dec. 2025 – Present
Undergraduate Researcher (Advisor: Hyoseok Hwang)

- Demonstrated that temporal regularization alone can induce spatial smoothness through shared next states, allowing policies to retain performance by avoiding explicit spatial regularization.
- Demonstrated that catastrophic forgetting in the RND predictor network can be harnessed to promote re-exploration of long-unvisited states, allowing effective adaptation to changing environments.

## PUBLICATIONS

Re-exploration by Leveraging Forgetting in Random Network Distillation
**SungJae Ahn**, Hyoseok Hwang
*Submitted to Korea Software Congress 2026*

Revisiting Temporal Regularization for Smooth Control in Deep Reinforcement Learning
**SungJae Ahn**, Jeong Woon Lee, Kyoleen Kwak, and Hyoseok Hwang
*Submitted to ICRA 2027*

## PROJECTS

**Mental Model Research for Robot Learning and Reasoning** Present
*Lab project with Korea Institute of Science and Technology (KIST)*

- Trained and evaluated $\pi_{0.5}$ and several memory-augmented VLA models on tasks that include long-horizon manipulation and occlusion.
- Investigating the use of privileged graphs during training to ground VLA models' memory updates in changes to objects and non-object elements in video.

**Analyzing Reasoning Diversity in Self-Distillation** Spring 2026
*Course project in Advanced Deep Learning*

- Demonstrated that leveraging feedback from failed rollouts can encourage diverse reasoning strategies in self-distillation policy optimization.
- Analyzed the distinct roles of KL divergence directions in failed and successful rollouts by relating their mode-covering and mode-seeking behavior to exploration and exploitation.

**Deepfake Detection Competition** Fall 2025
*Held by National Forensic Service*

- Ranked 33 out of 266 teams utilizing Dual Data Alignment for robust cross-domain forgery detection.
- Managed experiment configurations with Hydra and used W&B to track how dataset selection and preprocessing choices led to each training run.

## ACTIVITIES

**SW Mentor: AI 3001 Advanced Deep Learning** Present

- Designed programming assignments on core Transformer components.

**SW Mentor: CSE 304 Algorithms** Spring, Fall 2025

- Curated LeetCode problems for course exercises and supported students through online Q&A.

**KHUDA (KHU Data analysis and AI club) 7th** Jan. 2025 - Jul. 2025

- Built a legal-domain RAG chatbot to improve public access to Korean legal information, using Chroma vector search, Korean sentence embeddings, and MMR-based retrieval.

Signals Intelligence at Defense Security Agency Feb. 2022 - Nov. 2023

- Received awards for graduating first in my class (from a Major General) and identifying a previously unknown digital signal (from a Brigadier General).

## SKILLS

**AI/ML :** PyTorch, JAX, Gymnasium, Isaac Lab, Slurm

**English :** TOEFL 5.5 (110)

## AWARDS

**Academic Excellence Scholarship,** KHU Fall 2024 - Fall 2026